[INCH-POUND]
A-A-59590/1
November 27, 2000

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

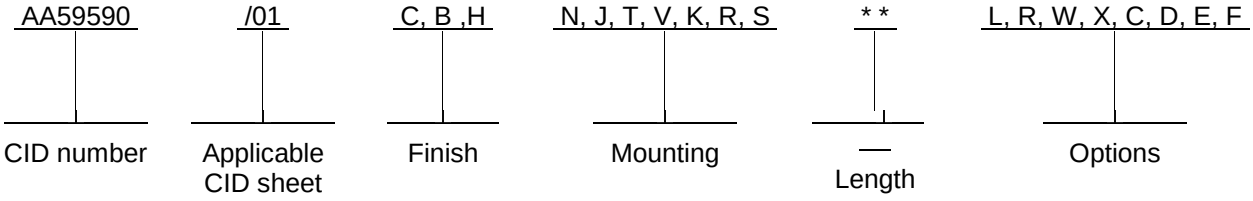
HOLDERS, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .225 X .260 INCH BODY SIZE, WITH LEVER ACTION LOCKING

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A-A-59590.

SCOPE. This CID specification sheet covers 3 piece card holders with a lever action locking feature that use screw actuated wedge retainers to hold circuit card assemblies into their installed positions in heat sinking devices (cold plates, heat exchanger) or other applications. Card holders covered by this CID specification sheet are intended to provide maximum contact between thermal paths on the circuit card assemblies and the heat sink surfaces and to provide resistance to shock and vibration. Items covered by this CID specification sheet are intended for commercial/industrial applications.

CLASSIFICATION. This CID specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see notes).



SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card firmly in place providing high resistance to shock and vibration while providing maximum thermal transfer.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein and the general CID.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center, Columbus, ATTN: DSCC/VAC, Post Office Box 3990, Columbus, OH 43216-5000, or facsimile (614) 693-6939, or electronic mail 5998_Documents@dscc.dla.mil.

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Lever. A lever-action device shall be used to secure the card holder and circuit card assembly in place. When in the relaxed state, the card holders shall allow the assembly to be placed into or removed from its application with a slight insertion/extraction drag. To secure the assembly in place, the lever shall be actuated, to become perpendicular with the card holder, and parallel with and towards the circuit card. The lever shall extend no more than 1.5 inches (38.10 mm) from the end of the card holder when the assembly is in the relaxed position. The lever shall extend no more than .40 inches (10.16 mm) from the end of the card holder when the assembly is in the expanded position.

Screw. The screw head and associated hardware shall not protrude more than .65 inches (16.51 mm) beyond the ends of the body and wedge assembly when the assembly is fully expanded.

Expanded and relaxed dimensions. Expanded and relaxed dimensions shall be as specified in figure 1.

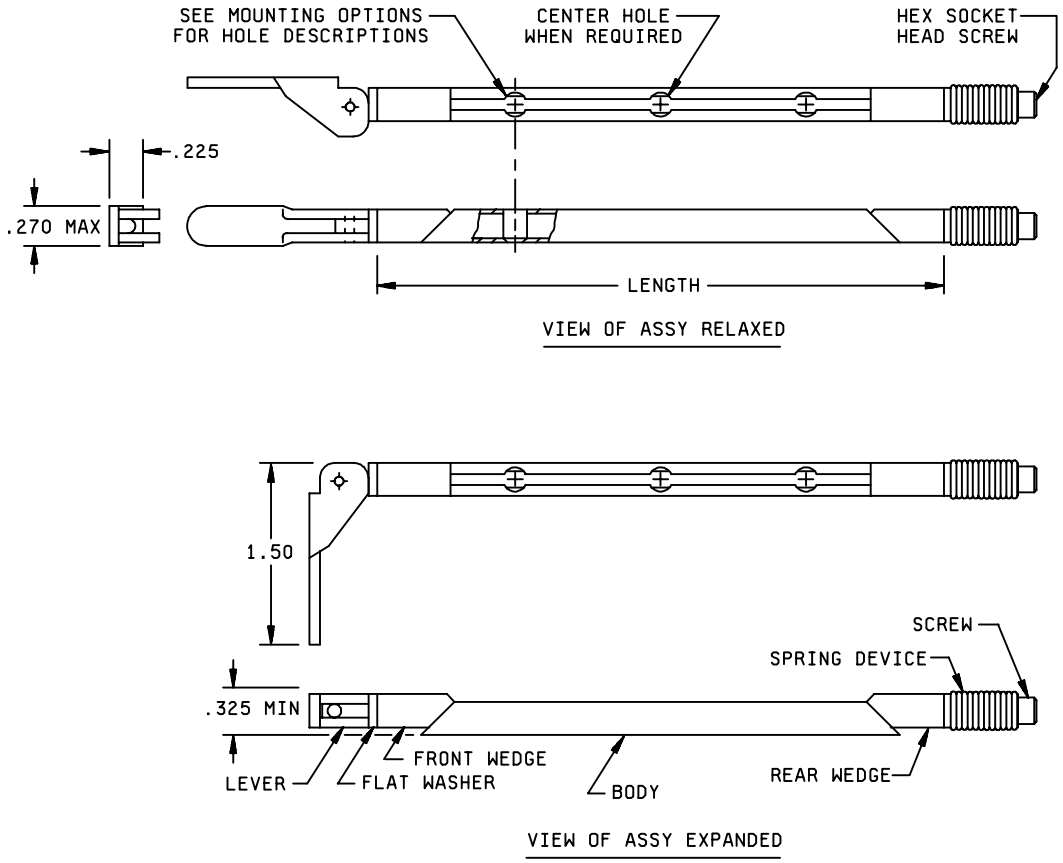

FIGURE 1 Relaxed and expanded dimensions. (Right hand configuration shown)

Mounting options. The mounting options shall be as specified in A-A-59590.

Mounting holes (when required). Two mounting holes are required on card holders less than 6.00 inches (152.4 mm) in length. The mounting holes shall be spaced using the length of the assembly minus 1.90 inches (48.26 mm) and centered on the mounting body. Three mounting holes are required on card holders 6.00 inches or greater in length. The third mounting hole shall be centered on the mounting body.

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Options. Card holders can have the following options. Card holders not requiring one of the following options shall include a suffix “N” in the PIN.

Facing (Left hand / Right hand). This card holder is available for mounting in either the left hand or right hand positions (see figure 2). Left facing card holders shall include a suffix “L” in the PIN. Right facing card holders shall include a suffix “R” in the PIN (see classification and notes).

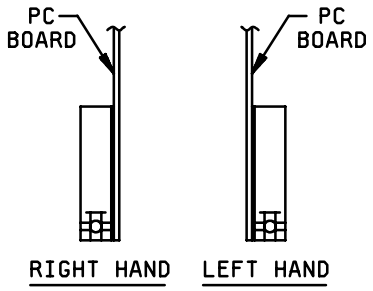

FIGURE 2 Facing Configuration.

Lockwasher and flatwasher. A lockwasher and flatwasher located under the screw head will provide for additional resistance to loosening from shock and vibration. Left facing card holders requiring a lockwasher and flatwasher option shall include a suffix “W” in the PIN (see classification and notes). Right facing card holders requiring a lockwasher and flatwasher option shall include a suffix “X” in the PIN (see classification and notes).

Screw self-locking element. The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock vibration. Left facing card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes). Right facing card holders requiring a screw self-locking element shall include a suffix “F” in the PIN (see classification and notes).

Lockwasher, flatwasher, and screw self-locking element. Left facing card holders requiring a lockwasher, flatwasher, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes). Right facing card holders requiring a lockwasher, flatwasher, and screw self-locking element option shall include a suffix “C” in the PIN (see classification and notes).

PRODUCT CONFORMANCE PROVISIONS. Product conformance shall be as specified in A-A-59590 and herein.

Performance tests. The following tests shall be performed with a 0.062 inch thick circuit card with a pair of 6 inch card holders using the manufacturer’s recommended clamping force adjustment procedures to secure the assembly to a suitable test fixture.

Vibration. Secure the card holder and circuit card assembly to a vibration system capable of producing an input of 25 Gs over a frequency range from 20 to 2000 to 20 Hz performed in 16 minutes. The assembly shall be able to withstand a vibration of 25 G-rms without dislodging or moving. This test shall be performed in both a perpendicular and parallel oscillation (in relationship to the circuit card) for a duration of 16 minutes in each orientation.

Shock. Secure the card holder and circuit card assembly to a standard drop shock tester capable of attaining at least 100 Gs. Subject the assembly to three shock pulses of 60g/6ms in each of the three principle planes (a total of 9 blows). The assembly shall be able to withstand the shock force without dislodging or moving.

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Thermal transfer. With the card holder and circuit card assembly secured to a suitable cold plate test fixture, the assembly shall have a 2 degrees to 4 degrees C/W/in thermal resistance when measured over a range from 1W to 90W, dissipated by conduction only.

NOTES

PIN. The PIN should be used for government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFGRs CAGE	MFGRs Name and Address	MGFRs contact information
18915	APW / Electronic Solutions 14100 Danielson Street Poway, CA 92064-6898	Tel: (858) 679-4550 Toll Free: (800) 854-7086 Fax: (858) 679-4555 E-mail: sales@apw-elsol.com URL: www.apw-elsol.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776-2021	Tel: (626) 287-0451 Fax: (626) 287-7350 E-mail: sales@calmark-corp.com URL: www.calmark-corp.com
71286	Fairchild Fasteners Camloc/Ram Products Division 3000 W. Lomita Blvd. Torrance, CA 90505-5103	Tel: (310) 784-2662 Fax: (310) 784-2628 E-mail: altedesco@fairchildfasteners.com URL: www.fairchildfasteners.com

Part number (P/N) supersession data. This CID specification sheet supersedes the following manufacturer's P/Ns as shown. This information is being provided to assist in reducing proliferation in the government inventory system.

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TABLE I. Commercial P/N supersession data.

PIN designator AA59590/01	Vendor similar designator or type part number 1/ 2/ CAGE 52094	CID PIN AA59590/01	Vendor similar designator or type part number 1/ 2/ CAGE 52094	CID PIN AA59590/01	Vendor similar designator or type part number 1/ 2/ CAGE 52094
BN28L	L225-2.80X	BV28L	L225-2.80	HT28L	LHA225-2.80T0
BN28R	L225-2.80X	BV28R	L225-2.80	HT28R	LHA225-2.80T0
BN30L	L225-3.00X	BV30L	L225-3.00	HT30L	LHA225-3.00T0
BN30R	L225-3.00X	BV30R	L225-3.00	HT30R	LHA225-3.00T0
BN35L	L225-3.50X	BV35L	L225-3.50	HT35L	LHA225-3.50T0
BN35R	L225-3.50X	BV35R	L225-3.50	HT35R	LHA225-3.50T0
BN38L	L225-3.80X	BV38L	L225-3.80	HT38L	LHA225-3.80T0
BN38R	L225-3.80X	BV38R	L225-3.80	HT38R	LHA225-3.80T0
BN48L	L225-4.80X	BV48L	L225-4.80	HT48L	LHA225-4.80T0
BN48R	L225-4.80X	BV48R	L225-4.80	HT48R	LHA225-4.80T0
BN53L	L225-5.30X	BV53L	L225-5.30	HT53L	LHA225-5.30T0
BN53R	L225-5.30X	BV53R	L225-5.30	HT53R	LHA225-5.30T0
BN55L	L225-5.50X	BV55L	L225-5.50	HT55L	LHA225-5.50T0
BN55R	L225-5.50X	BV55R	L225-5.50	HT55R	LHA225-5.50T0
BN58L	L225-5.80X	BV58L	L225-5.80	HT58L	LHA225-5.80T0
BN58R	L225-5.80X	BV58R	L225-5.80	HT58R	LHA225-5.80T0
BN61L	L225-6.10X	BS61L	L225-6.10ET2	HR61L	LHA225-6.10ET0
BN61R	L225-6.10X	BS61R	L225-6.10ET2	HR61R	LHA225-6.10ET0
BN67L	L225-6.70X	BS67L	L225-6.70ET2	HR67L	LHA225-6.70ET0
BN67R	L225-6.70X	BS67R	L225-6.70ET2	HR67R	LHA225-6.70ET0
BT28L	L225-2.80T0	HN28L	LHA225-2.80X	HV28L	LHA225-2.80
BT28R	L225-2.80T0	HN28R	LHA225-2.80X	HV28R	LHA225-2.80
BT30L	L225-3.00T0	HN30L	LHA225-3.00X	HV30L	LHA225-3.00
BT30R	L225-3.00T0	HN30R	LHA225-3.00X	HV30R	LHA225-3.00
BT35L	L225-3.50T0	HN35L	LHA225-3.50X	HV35L	LHA225-3.50
BT35R	L225-3.50T0	HN35R	LHA225-3.50X	HV35R	LHA225-3.50
BT38L	L225-3.80T0	HN38L	LHA225-3.80X	HV38L	LHA225-3.80
BT38R	L225-3.80T0	HN38R	LHA225-3.80X	HV38R	LHA225-3.80
BT48L	L225-4.80T0	HN48L	LHA225-4.80X	HV48L	LHA225-4.80
BT48R	L225-4.80T0	HN48R	LHA225-4.80X	HV48R	LHA225-4.80
BT53L	L225-5.30T0	HN53L	LHA225-5.30X	HV53L	LHA225-5.30
BT53R	L225-5.30T0	HN53R	LHA225-5.30X	HV53R	LHA225-5.30

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BT55L	L225-5.50T0	HN55L	LHA225-5.50X	HV55L	LHA225-5.50
BT55R	L225-5.50T0	HN55R	LHA225-5.50X	HV55R	LHA225-5.50
BT58L	L225-5.80T0	HN58L	LHA225-5.80X	HV58L	LHA225-5.80
BT58R	L225-5.80T0	HN58R	LHA225-5.80X	HV58R	LHA225-5.80
BR61L	L225-6.10ETO	HN61L	LHA225-6.10X	HS61L	LHA225-6.10ET2
BR61R	L225-6.10ETO	HN61R	LHA225-6.10X	HS61R	LHA225-6.10ET2
BR67L	L225-6.70ETO	HN67L	LHA225-6.70X	HS67L	LHA225-6.70ET2
BR67R	L225-6.70ETO	HN67R	LHA225-6.70X	HS67R	LHA225-6.70ET2

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph.

2/ Calmark (CAGE 52094) parts are furnished assembled in one direction. User must rotate the lever around for use in opposite direction.

MILITARY INTERESTS:

Custodians:
Air Force - 11
Army - CR
DLA - CC
Navy - EC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 7FXE

Preparing Activity

DLA - CC

Project 5998-0097-01

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